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Originator:	Talin Abu Gharbieh
Email:	talinee.ag@gmail.com
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01 - Methods of Measurement

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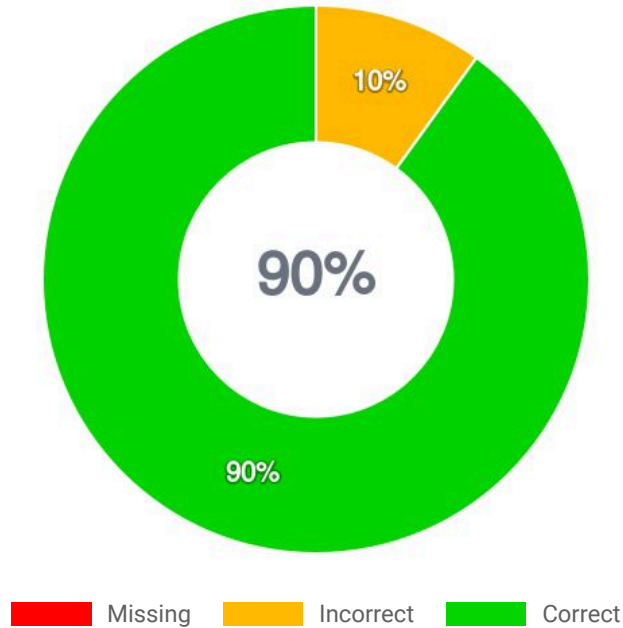
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1 Dashboards

Information Requirement Completions

Complete and Correct, Incomplete and Errors, Missing Requirements



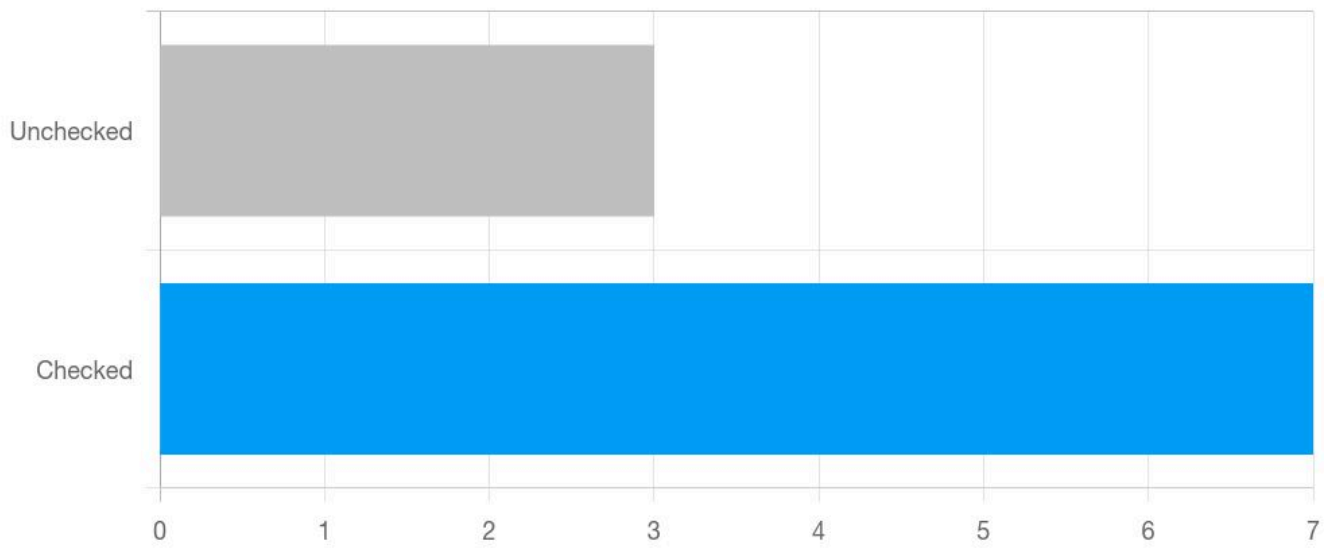
Scope Task Statuses

Pending, Proposed, Approved, Started, Complete, Verified



Scope Task Checklists

Unchecked vs Checked



2 Methods

2.1 Architecture

2.1.1 Interior Elements: Partition Walls, and Flooring Systems

Considering the numerous elements involved in the architectural environment, these complementing suggestions will specifically focus on Interior Partition walls and Flooring systems.

2.1.1.1 On-site Construction

Considering the architectural scope requirements for Milestone 1, this section establishes the guidelines for the on-site constructed solutions, specifically focusing on **Gypsum Board Partition Systems (Ss_25_15_35)** and **Flooring and Decking Systems (Pr_30_59_97)**.

A. Quantification and Modeling Guidelines

Regardless of the execution method, when quantifying these elements and their associated finishes using Building Information Models (BIM), it is imperative to use the net square area (m^2 or ft^2) as the standard quantification unit. All volumetric voids and penetrations must be accurately accounted for in the native model to ensure precise material takeoffs.

During early estimate strategies, quantities can be derived using the *Room* information parameters to approximate walls, floors, ceilings, and finishes, or by developing a model using multi-layered objects and applying average unit rates to the composite assembly.

For more accurate and detailed quantification during advanced project stages, it is essential to model partitions, floors, and their corresponding finishes as distinct, single-layered objects. This separation method allows for granular control over the modeling considerations specific to each material (e.g., differing installation heights, baseboards, or ceiling interactions).

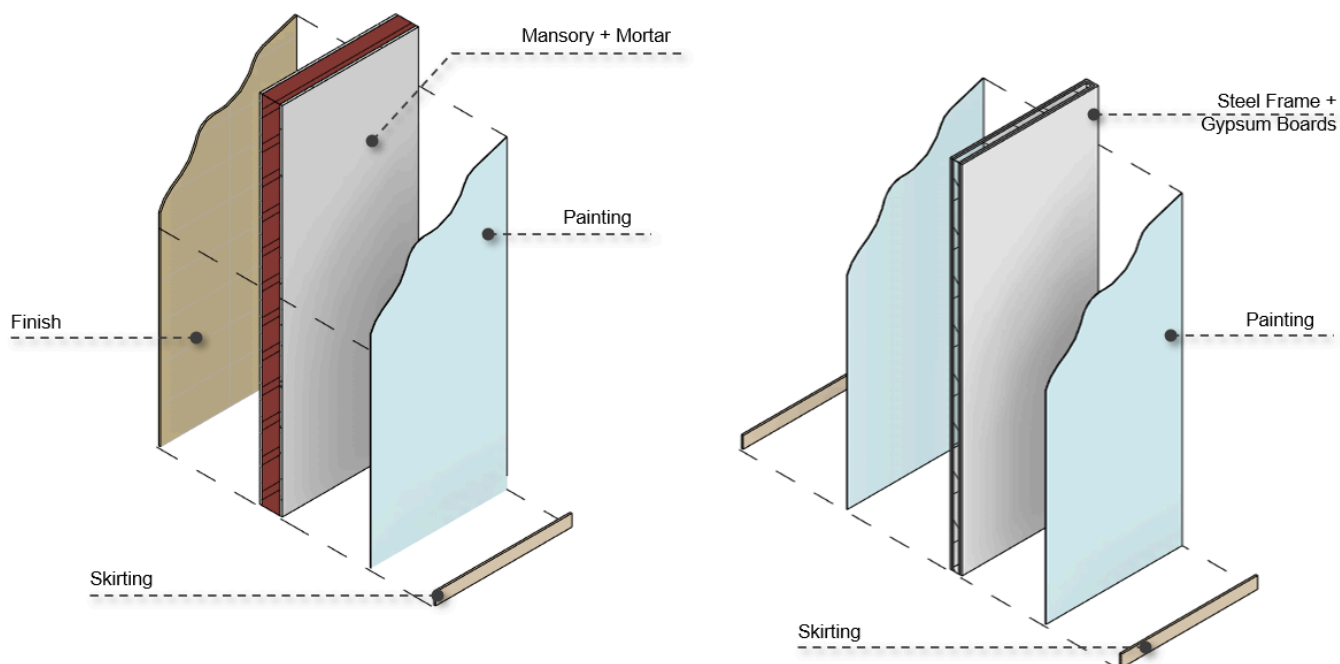


Figure 1: Figure 07 - Example of walls modelled as single separate-layered objects

Composite Construction Exceptions In this approach, each material is represented by an independent element, differentiating the core structure from applied finishes. However, for highly standardized composite elements—such as a drywall steel frame with gypsum plasterboards—modeling them as a consolidated layered element is acceptable, provided finishes (like paint or tiling) remain distinct.

- **Example A (Wet Construction):** A masonry wall element (mortar + masonry + mortar) with a separate tiling finish element. The tiling finish may be modeled to extend 10cm above the ceiling line.
- **Example B (Dry Construction):** A drywall element (gypsum board + steel frame + gypsum board) with a separate paint finish element extending up to the ceiling line.

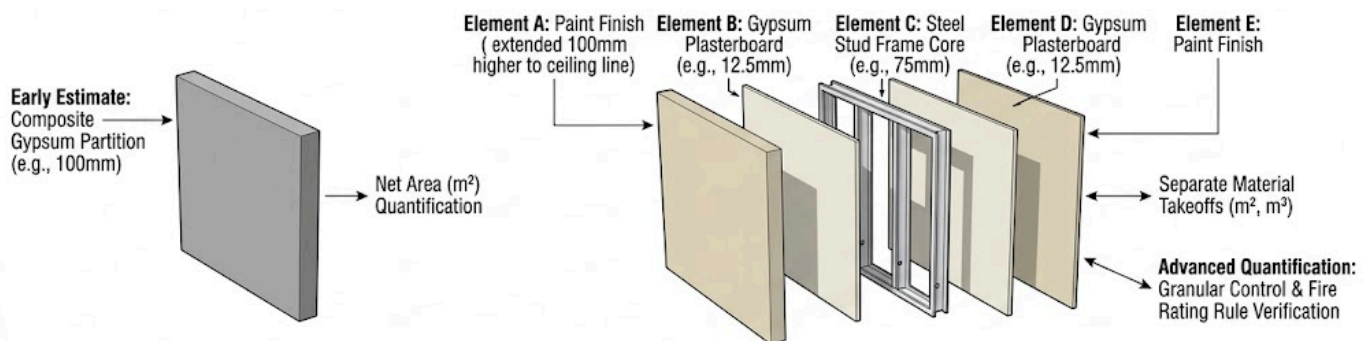


Figure 08 – Demonstration of Separated-Layer Modeling for Gypsum Partitions (Ss_25_15_35)

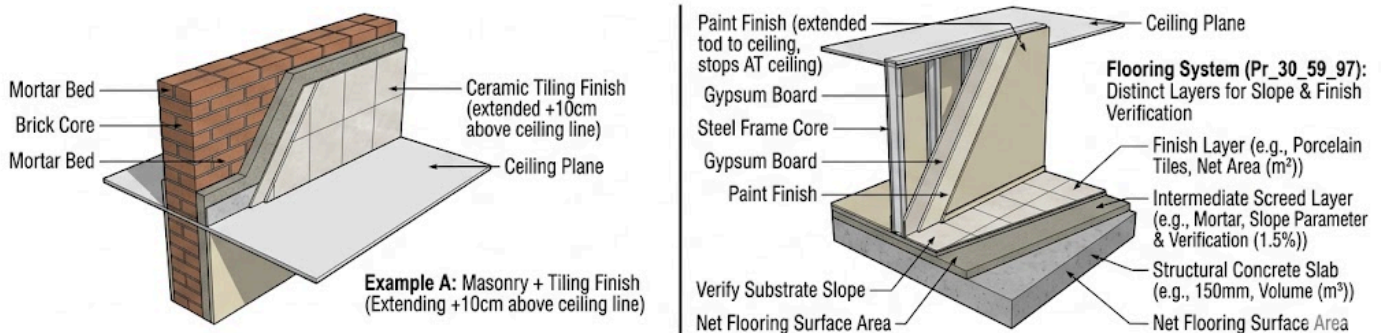


Figure 09 – Examples of Separated-Layer Walls and Floors, Considering Constructive Height Needs (Ss_25_15_35 & Pr_30_59_97)

Figure 2: Figure 08 - Example of walls modelled as single separate-layered objects. Figure 09 – Examples of single separate-layered walls, considering the height according to each constructive need of each material.

This separation method allows better control over the modeling considerations specific to each material, resulting in more accurate quantities.

B. Milestone 1: Scope & Information Requirements

To ensure BIM compliance and facilitate automated model checking, the following parameters must be strictly populated for the interior partition and flooring elements. Fire rating data is particularly critical and will be verified using automated rule-based compliance engines.

Element Scope	Classification	Required Parameters (Milestone 1)	Validation / Rule
Gypsum Board Partition Systems	Ss_25_15_35	• Area • Length • Width • Volume • Absorptance • Roughness • Type Name • Fire Rating	Must exclude empty values. Fire Rating Rule: <code>^[A-Z]{2}[1,5]?/[A-Z,3]/[S]</code>
Flooring and Decking Systems	Pr_30_59_97	• Area • Volume • Thickness • Slope • Roughness • Thermal Resistance (R) • Type Name	Must exclude empty values. Quantities must reflect net values.

C. On-Site Measurement & Verification

The AEC industry requires precise measurement methods to ensure execution aligns with the digital twin. Below are the standard practices and required tooling for on-site construction of partitions and flooring systems.

Adhering to **accurate measurement techniques** is crucial for project efficiency and quality. Below are the steps involved in on-site construction measurements:

1. Identify the precise measurement requirements based on the LOD/LOI scope parameters.
2. Select appropriate digital or analog tools for the task.
3. Conduct initial on-site measurements and record data directly against model elements.
4. Verify physical measurements against the BIM model for acceptable tolerances.
5. Generate non-compliance reports or update the as-built model based on verified data.

There are several benefits to using precise measurement methods during on-site construction:

- Enhanced project accuracy
- Reduced material waste
- Improved time management
- Cost efficiency

By employing these methods, the **AEC industry** can ensure that construction meets design specifications and quality standards.

Measurement Type	Required Tools	On-Site Purpose
Linear Measurement	Tape Measure, Laser Measure	Verifying structural track lengths, stud spacing, and partition heights.
Area & Level Measurement	Laser Level, Measuring Wheel	Calculating net flooring surface areas and ensuring substrate slope compliance.
Volume / Spatial Validation	3D Laser Scanner (LiDAR)	Capturing point clouds of completed assemblies to verify as-built models and clash detection.

🌐 2.1.1.2 Pre-fabricated Construction

Pre-fabricated (or modular) solutions should be modeled and quantified according to their system particularities, either by area or count by type.

In this scenario, the recommended approach is to develop the model using multi-layered objects and apply the pricing to this composition.

🌐 2.1.1.3 Insulation

The quantification methods will vary as follows:

- If the insulation is modeled, it should be quantified according to its construction methodology, such as length, area, or volume.
- If the insulation is not modeled, it should be included in the composition of the host architectural / structural element and described within the corresponding Bill of Quantities (BoQ) item.

🌐 2.1.2 Stair Finishes

Stairs finish, however, exhibit a different behaviour compared to the elements previously mentioned.

When quantifying stair finishes, considering the limitations present in software like Revit, it's important to note that they cannot be automatically quantified when embedded into stairs.

To address this challenge, the following methods can be employed:

🌐 2.1.2.1 Derived Takeoff

Utilize the dimensions of the stairs to calculate the finish area.

Depending on the finish type, the quantification unit should be either length (m or ft) or square area (m² or ft²).

- Finishes Quantified by Area:

$$\text{Threads Finish Area (m}^2\text{)} = \text{number of threads} * \text{width} * \text{length}$$

$$\text{Risers Finish Area (m}^2\text{)} = \text{number of risers} * \text{height} * \text{length}$$

$$\text{Stair Finish Area (m}^2\text{)} = \text{Threads Finish Area} + \text{Risers Finish Area}$$

- Finishes Quantified by Length:

$$\text{Threads Finish Length (m)} = \text{number of threads} * \text{length}$$

$$\text{Risers Finish Length (m)} = \text{number of risers} * \text{length}$$

$$\text{Stair Finish Length (m)} = \text{Threads Finish Length} + \text{Risers Finish Length}$$

🌐 2.1.2.2 Material Takeoff

Apply the designated finish material to the treads and risers of the modelled stairs.

2.1.3 Carpentry, Furniture, Equipment, Doors, and Windows

When quantifying carpentry, equipment, furniture, doors, and windows using BIM, it's crucial to assemble these elements exactly as they will be quantified, aligning with the specific requirements of the project.

The quantification is unitary, therefore, the modeled components should be grouped into a single element to form a distinct BoQ item, as shown in Figure 10.

Material takeoff is only employed by the subcontractor to price each item individually.

Dealing with repetitive elements like doors, windows, and plumbing fixtures demands a consistent approach.

To ensure precise quantification of multiple instances of the same BoQ item, the code must remain the same, as demonstrated in Figure 11.

It's worth noting that if there are any variations among objects under the same code, such as differences in description or specification, the code should be modified to ensure precise cost estimation aligned with each element's specification.

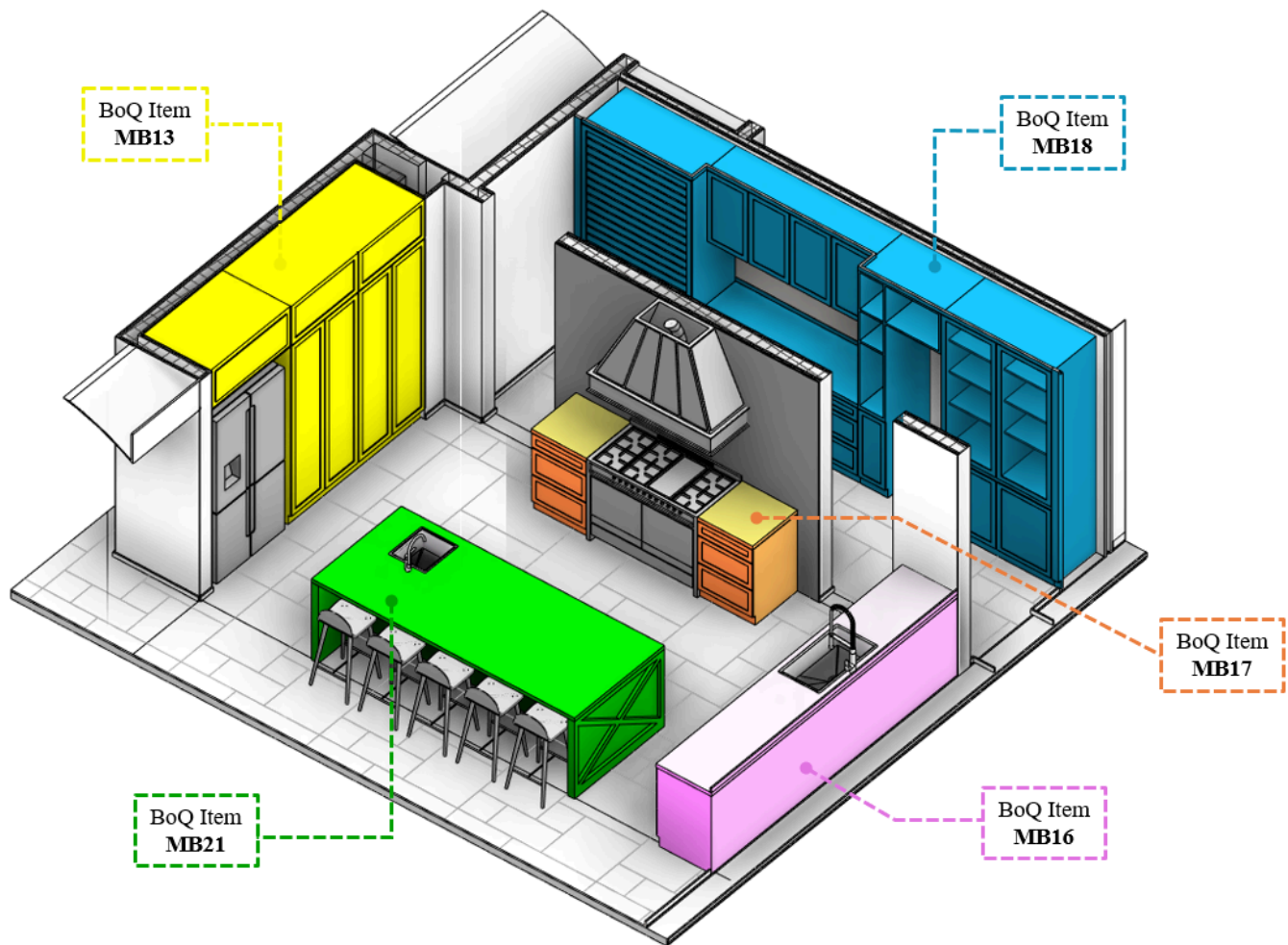


Figure 10 - Illustration of casework model assembly, where each colored casework assembly represent one BoQ item.

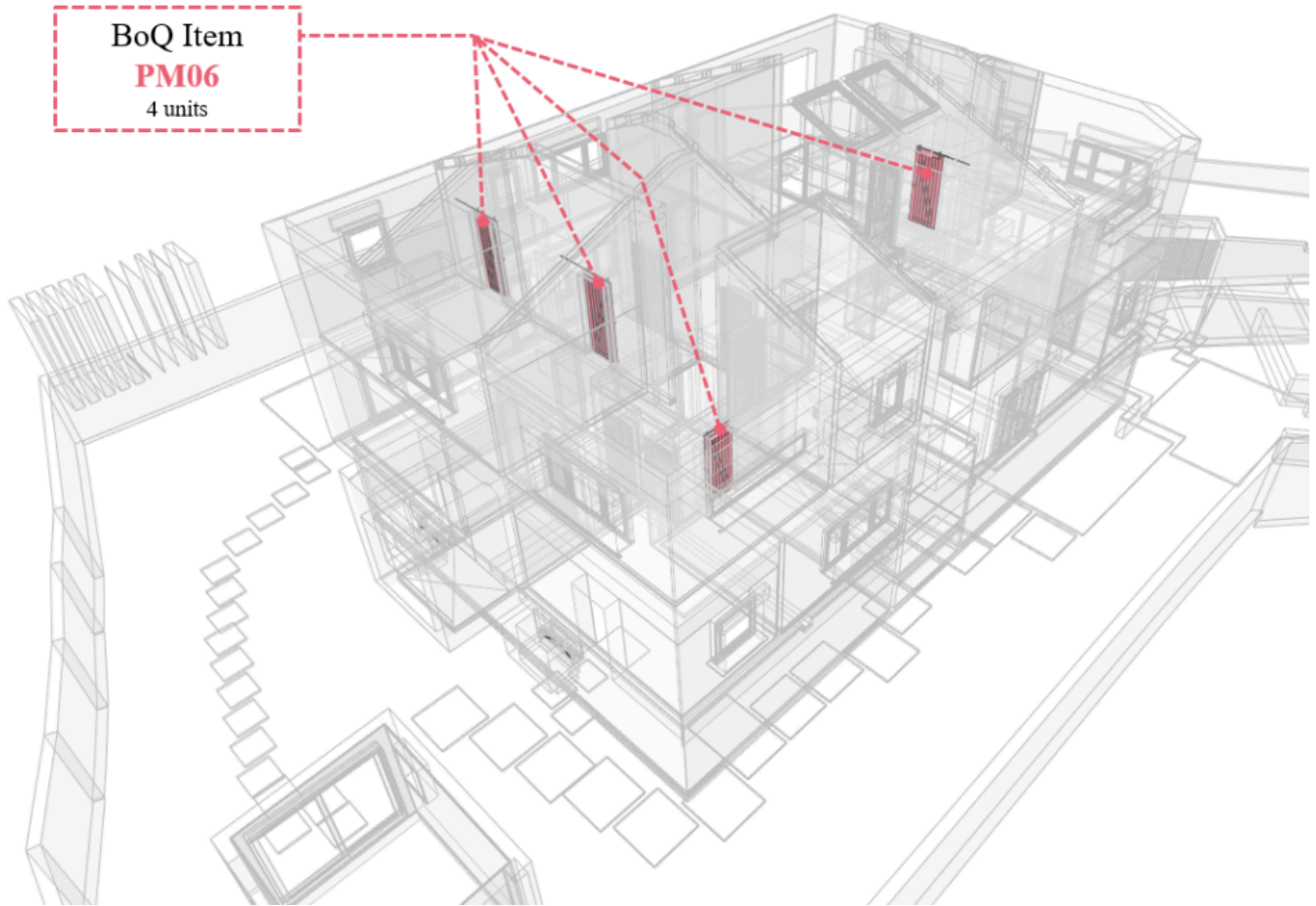


Figure 11 - Illustration of repetitive BoQ items with the same code, in this case the door PM06.

2 - Scope

		Milestone 1	
		May-10-2026 - Jun-10-2026	
Ss_25_13_35 Gypsum board partition systems IfcWall	Detailed/Proposed	 Verified	ABC 1
	Absorptance	✓	
	Area	✓	
	Fire Rating	✓	
	Length	✓	
	Roughness	✓	
	Type Name	✓	
	Volume	✓	
	Width	✓	
Pr_30_59_97 Flooring and decking systems OST_Floors	Detailed/Proposed	 Verified	ABC 2
	Area	✓	
	Roughness	✓	
	Slope	✓	
	Thermal Resistance (R)	✓	
	Thickness	✓	
	Type Name	✓	
	Volume	✓	

3 - Timeline



4 - Scope Details

Milestone: Milestone 1

**1 Ss_25_13_35 Gypsum board partition systems**

IfcWall

TRACKING

ABC: My Team
Unassigned
18 May 2026 - 27 May 2026

Verified

GEOMETRY**Detailed/Proposed**

Geometric extents and shapes necessary to ensure that later modeled components will fit around and within the space available, integrated with major nearby or attached elements.

DESCRIPTION

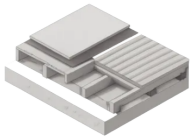
This task involves the creation and detailing of gypsum board partition systems for the project. The geometry level required is "Detailed", ensuring precise specifications and dimensions are adhered to. This task is currently at the "Proposed" reliability stage and is scheduled for completion by "Milestone 1".

CHECKLIST

- ☒ Verify gypsum board specifications
- ☒ Ensure detailed geometry is modeled accurately
- ☒ Review proposed design for compliance with standards
- ☐ Coordinate with structural and MEP teams
- ☐ Conduct quality check before finalizing Milestone 1

INFORMATION REQUIREMENTS

- ☒ Absorptance
- ☒ Area
- ☐ Fire Rating
- ☒ Length
- ☒ Roughness
- ☒ Type Name
- ☒ Volume
- ☒ Width

**2 Pr_30_59_97 Flooring and decking systems**

OST_Floors

TRACKING

ABC: My Team
Unassigned
11 May 2026 - 29 May 2026

Verified

GEOMETRY**Detailed/Proposed**

Geometric extents and shapes necessary to ensure that later modeled components will fit around and within the space available, integrated with major nearby or attached elements.

DESCRIPTION

This task involves the detailed design and specification of flooring and decking systems for the project. The geometry level is set to "Detailed," ensuring comprehensive modeling of all components. The reliability is currently marked as "Proposed," indicating that it is subject to further review and confirmation. Completion of this task is targeted for "Milestone 1."

CHECKLIST

- ✓ Review detailed flooring and decking system specifications
- ✓ Verify material selections and quantities for flooring and decking
- ✓ Coordinate with structural engineer for load requirements
- ✓ Ensure compliance with building codes and standards
- Prepare installation schedule aligned with Milestone 1

INFORMATION REQUIREMENTS

- ✓ Area
- ✓ Roughness
- ✓ Slope
- ✓ Thermal Resistance (R)
- ✓ Thickness
- ✓ Type Name
- ✓ Volume

5 - Information Requirements

Milestone 1
May-10-2026 - Jun-10-2026

			1. Absorbance	2. Area	3. Fire Rating	4. Length	5. Roughness	6. Slope	7. Thermal Resistance (R)	8. Thickness	9. Type Name	10. Volume	11. Width
Ss_25_13_35	Gypsum board partition systems	ABC	✓	✓	✓	✓	✓				✓	✓	✓
Pr_30_59_97	Flooring and decking systems	ABC		✓			✓	✓	✓	✓	✓	✓	

6 - Information Legend

Key	Type	Name (Group)	Description	Verification Rule	Value	Units	Tags
1	IFC	Absorptance		Any value (excluding empty)			
2	IFC	Area	The area of the tiled floors	Any number			
3	IFC	Fire Rating	Fire rating	Define Rule (Using AI)	$^{\wedge}([A-Z]\{2\})\backslash s?(\backslash d\{2,3\})\$$		
4	IFC	Length	[[Length]] of ceiling tiles. The size information is provided in addition to the shape representation and the geometric parameters used within. In cases of inconsistency between the geometric parameters and the size properties, provided in the attached property set, the geometric parameters take precedence.	Any value (excluding empty)			
5	IFC	Roughness		Any value (excluding empty)			
6	IFC	Slope		Any value (excluding empty)			
7	IFC	Thermal Resistance (R)	Has Thermal Resistance properties	Any value (excluding empty)			
8	IFC	Thickness		Any value (excluding empty)			
9	IFC	Type Name		Any value (excluding empty)			
10	IFC	Volume		Any value (excluding empty)			

Key	Type	Name (Group)	Description	Verification Rule	Value	Units	Tags
11	 IFC	Width		Any value (excluding empty)			